

Global composite drought indices for investigating the effects of climate-driven drought on major crop water requirements

Muhammad Abrar FAIZ^{1†}, Liangliang ZHANG^{1†}, Dong LIU^{1,2,3,4*}, Yongqiang ZHANG^{5*}, Mo LI¹ & Farah NAZ⁶

¹ School of Water Conservancy & Civil Engineering, Northeast Agricultural University, Harbin 150030, China

² Joint International Research Laboratory of Habitat Health of Black Soil in Cold Regions, Ministry of Education, Northeast Agricultural University, Harbin 150030, China

³ Key Laboratory of Effective Utilization of Agricultural Water Resources of Ministry of Agriculture and Rural Affairs, Northeast Agricultural University, Harbin 150030, China

⁴ Heilongjiang Provincial Key Laboratory of Water Resources and Water Conservancy Engineering in Cold Region, Northeast Agricultural University, Harbin 150030, China

⁵ Key Laboratory of Water Cycle and Related Land Surface Processes, Institute of Geographic Sciences and Natural Resources Research, Chinese Academy of Sciences, Beijing 100101, China

⁶ Department of Civil Engineering, Khwaja Fareed University of Engineering & Information Technology, Rahim Yar Khan 64200, Pakistan

* Corresponding authors: Dong LIU (liudong@neau.edu.cn), Yongqiang ZHANG (zhangyq@igsrr.ac.cn)

† These authors contributed equally to this work.

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Abstract Droughts are expected to increase in warmer climates, significantly affecting global crop production and raising crop water demand. The recurrence of extreme droughts in recent years needs more attention to capture their multifaceted nature. Explaining the impact of general drought indices on water requirements could limit the precision of future water demand estimates. Three composite drought indices were used to investigate the effects of drought on crop water requirement (CWR), irrigation water requirement (IWR), and yields for major crops under the Shared Socioeconomic Pathways (SSPs 245 and 585), to provide a broader view of future water-demand risks. The results indicate that drought distribution is accompanied by wetness, with a higher likelihood of future dryness in regions such as South Asia, Australia, and North and South America. There is an increase in the estimates of the CWRs and IWRs under the SSPs. In extreme cases (SSP 585), maximum fractional changes in total (including rainfed and irrigated) CWR and IWR relative to the baseline period are +80.16 mm/month and +42.68 mm/month for wheat; +52.54 mm/month and +7.09 mm/month for maize; and +14.01 mm/month and +4.98 mm/month for rice, respectively. Overall, the results indicate a significant increase in CWR and IWR under future climate scenarios, underscoring the potential for increased water demand and the urgent need to develop adaptive water management strategies.

Keywords Drought, Evapotranspiration, Major crops, Monitoring, CMIP6

1. Introduction

Growing evidence indicates the intensification of climate extremes (Xu et al., 2019; Seneviratne et al., 2021), especially the land area at risk from drought under projected climate models (Harrison et al., 2014). Drought is the most influential and widespread natural disaster and a significant threat to agricultural crop production (Liu et al., 2016). Agriculture is the primary source of income in many countries, and approximately 1.3 billion people rely on it; this sector is highly affected by drought (Pourzand and Noy, 2022). In the last two decades, many countries have experienced substantial drought losses (Attiwill and Adams, 2013; Basara et al., 2019). A large portion of freshwater is used by agriculture, and the availability of water resources is at risk because of rapid population growth and climate change (Zhang et al., 2023). Increases in temperature under climate change accelerate global hydrological cycles and cause variations in the spatial distribution of global droughts (Naumann et al., 2018). Thus, global spatial dryness and its effect on agricultural crop water requirements via updated drought indices are crucial for mitigating adverse effects on food production, agriculture, and natural resources.

Maize, rice, and wheat are essential cereals that guarantee feed and food security worldwide (Shiferaw et al., 2013; Challinor et al., 2014). However, their growth and productivity are tied to meteorological factors like drought stress. Precipitation is a key climate variable that mostly meets crop water requirement (CWR), and crop yield production is vulnerable to drought due to the variability of effective precipitation and high temperatures (Naresh Kumar et al., 2014; Wang J et al., 2018a). The increasing risk of drought and the decrease in regional projected precipitation may further cause a decline in the production of these cereals (O'Leary et al., 2018; Wang J et al., 2018b). Water stress triggers yield loss if irrigation water shortages occur at any crop growth stage (Anjum et al., 2017; Dietz et al., 2021). Optimizing irrigation schemes and extending the irrigation network could help mitigate the adverse impacts of drought on crop production. However, only 25% of the area is irrigated worldwide based on MIRCA2000, and a portion of dominant crops, such as maize (0.3 million km²), wheat (0.7 million km²), and rice (1.0 million km²) is harvested in irrigated areas. Maize and wheat are largely grown in 1.2 million km² to 1.5 million km² rainfed harvest areas (Portmann et al., 2010). Seasonal drought can delay the sowing of crops, influencing grain yield

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(Dietz et al., 2021). Thus, it is necessary to understand the CWR and irrigation water requirement (IWR) distributions during drought to better manage water shortages and design irrigation schemes (Kraemer, 2015; Boonwichai et al., 2018).

Several drought indices have been developed to monitor and quantify drought at temporal and spatial scales (Dai, 2011; Zargar et al., 2011; Faiz et al., 2021). Among these, the Standardized Precipitation Index (SPI), Standardized Precipitation Evapotranspiration Index (SPEI), Palmer Drought Severity Index (PDSI), Crop Moisture Deficit Index (CMDI), and terrestrial water storage metrics (integration of snow, groundwater, and soil moisture deficit) are used for drought characterization because they effectively represent different drought types (Palmer, 1965; McKee et al., 1993; Narasimhan and Srinivasan, 2005; Vicente-Serrano et al., 2010; Pokhrel et al., 2021; Yin et al., 2022). Likewise, researchers developed composite and multivariate drought indices to track the progression from meteorological deficits to soil-moisture stress and their impacts. For example, the composite/integrated or combined drought indicators, which integrate precipitation, soil-moisture anomaly, and vegetation stress to monitor and alert to the drought and its impact on reduced plant activity (Sepulcre-Canto et al., 2012). The Multivariate Standardized Drought Index (MSDI) pairs precipitation with soil moisture index (Hao and AghaKouchak 2013), while the Joint Deficit Index (JDI) uses a copula framework to link joint precipitation and streamflow deficits to drought probabilities (Kao and Govindaraju, 2010) and Integrated Drought Condition Index (IDCI) used meteorological drought index, vegetation conditions and soil moisture to assess the agricultural drought hazards (Shen et al., 2019). These indices collectively enhance our understanding of drought dynamics by providing insights into the severity, duration, and spatial extent of drought events. They are widely used but have limitations due to their methodological differences and varying effectiveness in different regions. A drought index should be statistically robust and capable of comparing performance across diverse areas (Keyantash and Dracup, 2002). To address these limitations, composite drought indices that combine various hydro-climatic factors have been developed, such as the Composite Drought Index (CDI) (Faiz et al., 2022). The CDI integrates a modified rainfall anomaly index (RAI_m), a water-balance anomaly index (WBAI), and a moisture index (MI, calculated from precipitation and actual evapotranspiration). By considering rainfall deficits, water balance, and evaporative demand, the CDI provides a comprehensive view of drought processes. During its development, the CDI was tested against soil moisture, streamflow, and scaled crop yield indices across China, showing strong correlations with agricultural impacts and fewer false alarms compared to PDSI and evaluation integrated drought index (EIDI) (Faiz et al., 2022). Recently, CDI has been enhanced with different evapotranspiration formulation datasets and machine learning techniques, further aligning it with soil moisture and crop water requirements (Faiz et al., 2025). These results showed that the CDI is both scientifically robust for assessing agricultural water stress. However, our goal is not to suggest that the CDI is universally better or that it replaces existing drought monitoring systems. Instead, we use the CDI as a tool to generate a consistent drought signal that can be directly connected to CWR and IWR under future climate scenarios. Although numerous multivariate indices exist (Hao and Agha-Kouchak, 2013; Zargar et al., 2011), few studies have directly

linked these indices to agricultural water demand and irrigation planning (Wada et al., 2013; Nechifor and Winning, 2019). The novelty of our work lies in combining the three sets of CDIs with CWR/IWR estimates and multi-model climate projections, enabling us to measure how changing drought patterns will impact global crop water demands. This approach connects drought diagnosis with resource management. Moreover, many studies have reported that drought severity may increase under a warming climate, as reflected by increased stress on water resources and vegetation, increased drought risk, crop IWRs, and CWRs (Murray et al., 2012; Wang Z et al., 2018; Vicente-Serrano et al., 2020). Droughts substantially affect crop yields and agricultural productivity by altering CWR and IWR, resulting in lower agricultural output and economic losses (Lobell et al., 2011; Daryanto et al., 2017; Leng and Hall, 2019). Thus, applying the CDIs is crucial for understanding the effects of drought on IWR and CWR, as well as the future implications for agricultural water management, irrigation scheduling, and irrigation infrastructure planning. In other words, it enhances the ability to improve crop productivity and water use efficiency.

This study aims to examine the relationships between drought and CWR and IWR and provide a context for water requirements at a global scale to inform irrigation water resource decision-making. Particularly, we set up the following objectives: (1) To analyze the spatial distribution of drought conditions and their relationships with CWR and IWR; (2) To project future changes in drought, CWR, and IWR under multiple climate scenarios; (3) To assess the CWR and IWR for historical periods and future scenarios in different cropping areas.

2. Data and methods

2.1 Data sources

Monthly precipitation and actual evaporation (AE) (extracted from global land evaporation Amsterdam mode (GLEAM)) and evapotranspiration datasets were collected from online sources from 2000 to 2021 and from 2024 to 2099. Historical precipitation data were obtained from multisource weather (MSWX) (www.gloh2o.org/mswx) (Beck et al., 2022). MSWX is a global high-resolution data product available over all land and oceans. This dataset is widely used in climate analysis, drought monitoring, and hydrological studies (Araghi et al., 2023). For future periods, precipitation data were obtained under shared socioeconomic pathway (SSP) scenarios, including SSP245 and SSP585 (<https://cds.climate.copernicus.eu/cdsapp#!/dataset/projections-cmip6?tab=form>) (Copernicus 2021). The GLEAM AE (Martens et al., 2017) was obtained for the historical period, whereas for future periods, potential evapotranspiration (PET) was obtained under SSP245 and SSP585 from <https://doi.org/10.5281/zenodo.7789759> (Bjarke et al., 2023). The selection of SSP245 was made based on the hypothesis that medium greenhouse gas emissions and better mitigation efforts characterize it.

GLEAM AE is a summation of transpiration, bare soil evaporation, open-water evaporation, interception loss, and snow sublimation, whereas PET is based on the FAO-56 Penman-Monteith calculation method (Allen et al., 1998). All the datasets were regridded to 0.25° resolution via bilinear interpolation (see Appendix Tables S1 and S2 for dataset details). We used nine GCMs under three SSPs for the CDIs component

calculations. Major crop yield data were obtained from <https://doi.pangaea.de/10.1594/PANGAEA.909132> from 2000 to 2016 to assess the relationships between drought indices and annual yield. The spatial distribution of the yield data is presented in Appendix Fig. S1.

2.2 Composite drought index (CDI)

The *CDI* is a composite of the *WBAI*, *RAI_m*, and *MI* (Faiz et al., 2022). For the *WBAI* and *RAI_m*, the precipitation total and median water balance, which is based on the difference between precipitation and *AE/PET* (in this case), were used as the average distribution, and average wet and dry extreme cases were used to define the distribution variability (Hänsel et al., 2019). *RAI_m* captures the precipitation-driven meteorological signal; *WBAI* adds evaporative demand via the difference between precipitation and *AE/PET*, which is closely related to water-balance deficits; *MI* converts conditions to a moisture ratio linked to soil moisture and crop water stress. Empirically, the *CDIs* showed stronger positive correlations with gridded soil moisture and scaled crop yields than *PDSI/EIDI* in the regional evaluation (Faiz et al., 2022), and statistically significant ($r > 0.5$) correlations with *CWR* in the revised formulations of *CDI* (Faiz et al., 2025). The basis *CDI* is written as:

$$CDI = a \cdot RAI_m + b \cdot WBAI + c \cdot MI \quad (1)$$

where a (0.4), b (0.4), and c (0.4) values were taken from (Zou and Zhang, 2008) to balance the equation. These coefficients were used to test the *CDI* in Chinese mainland. China encompasses diverse agro-climatic zones (humid subtropical to arid/semi-arid and high-elevation), making it a valuable regional testbed without implying that it applies globally. Therefore, we assume that it may indicate similar dryness patterns worldwide, although further analysis with global datasets is needed to confirm its validity.

Two moisture formulations are used as an experiment in *CDI*:

(1) Historically referenced moisture (MI_{his}), which is the same as the original *CDI*:

$$MI_{his} = \left(\frac{P - AE \text{ or } PET}{P - AE \text{ or } PET} \right) \quad (2)$$

(2) Current moisture (MI_{curr}):

$$MI_{curr} = \left(\frac{P - AE \text{ or } PET}{P} \right) \quad (3)$$

Based on eqs. (2) and (3), the *CDIs* are:

$$CDI_W = a \cdot RAI_m + b \cdot WBAI + c \cdot MI_{his} \quad (4)$$

$$CDI_NCC = a \cdot RAI_m + b \cdot WBAI + c \cdot MI_{curr} \quad (5)$$

Eq. (5) aligns the weights with the National Climate Center's composite practice and uses MI_{curr} as in the revisiting experiment (Faiz et al., 2025). *CDI_NCC* is the whole linear combination. Additionally, we combined these indices using principal component analysis (PCA) to reduce dimensionality and facilitate comparison, thereby yielding more robust results. To combine these three indices (*WBAI*, *RAI_m*, and *MI*) into a single matrix, eq. (6), “*sklearn.decomposition.PCA*”, was used to perform PCA.

(3) *CDI_Main*

$$X = [RAI_m \quad WBAI \quad MI]^T \quad (6)$$

Eq. (6) represents the vector of three input matrices. Each component of X is standardized to obtain Z . Let $Cov(Z)$ be the covariance matrix of Z . u_1 is the leading eigenvector with its largest eigenvalue. The main composite index as the projection of Z :

$$CDI_Main = u_1^T \cdot Z \quad (7)$$

This construction is the first principal component of the three standardized inputs (i.e., applying PCA to X and retaining principal component 1 (PC1)). This efficiently avoids ad-hoc weights while retaining the dominant joint variability.

CDI_W (weighted: indicates the use of coefficient values to balance the equation), *CDI_NCC* (indicates the use of coefficient values to balance the equation and replace the *MI* with recent moisture calculations), and *CDI_Main* (combines the *WBAI*, *RAI_m*, and *MI* to make the *CDI* with PCA, in which the *MI* involves long-term precipitation and actual evaporation/evapotranspiration, as discussed above) were used. The classifications of *CDI_Main*, *CDI_W*, and *CDI_NCC* are presented in Appendix Table S3. For a detailed calculation of *RAI_m* and *WBAI*, refer to the supplementary material section (Appendix S1). In addition, the *SPI* and *SPEI* were also calculated via the same precipitation and potential evapotranspiration data to compare *CDI_Main*, *CDI_W*, and *CDI_NCC*. The *SPI* and *SPEI* procedures are also presented in the supplementary material. A one-month time scale is usually used to measure the short-term fluctuations in the drought and ensure continuity in the computation of the *SPI* and *SPEI* indices. The 1-month scale provides a sensitive measure of short-term anomalies in precipitation and atmospheric water demand, which are particularly relevant for detecting the onset of droughts and understanding their rapid progression. In this study, a 1-month scale was used to focus on early drought detection and to analyze short-term hydrometeorological variability that directly impacts crop water stress. This approach is particularly vital for understanding the effects of drought during critical crop growth stages, where even short-term water deficits can lead to significant agricultural consequences. Furthermore, the regional validation of *CDI_Main* has already been done against crop yield, streamflow, and gridded soil moisture. Please refer to the published material (Faiz et al., 2022).

2.3 Computation of CWRs and IWRs

The monthly *CWR* was computed based on the crop coefficient and *PET* via the FAO Penman-Monteith formula (Doorenbos and Pruitt, 1975).

$$CWR = ET_0 \times K_c \quad (8)$$

where K_c is a crop coefficient obtained from the Food and Agriculture Organization (FAO) for cash crops (maize, wheat, and rice). We assign K_c values according to crop growth stages in different geographic regions. Three crop growth stages (initial, middle, and maturity) were taken to assign crop coefficient values. A crop plantation month was checked in the FAO database to determine the growth stages during the year in different regions. For example, the maize crop planted in April in the US would be in its initial growth stage by June, but the same crop planted in Argentina in October would be in its middle stages by January. Understanding the differences in crop planting practices across various countries and regions, the *CWR* calculations were made to produce more accurate and reliable

estimates.

The *IWR* was calculated via the *CWR* and effective precipitation differences.

$$IWR_{t,i,j} = \max(0, CWR_{t,i,j} - Pe_{t,i,j}) \quad (9)$$

where *Pe* represents the effective precipitation at time *t*, longitude *j* and latitude *i*; and $\max(0, \cdot)$ prevents a zero *IWR* from being reached. For the *Pe* calculation, CROPWAT model equations were used (Smith, 1992). The equations are based on a fixed percentage of precipitation, as some of the precipitation could be lost due to percolation and runoff (evapotranspiration losses are accounted for separately in the *ET₀* and *CWR* calculations). In eq. (10), a 20% loss of precipitation was taken, while a 10% loss of precipitation is counted in equation 11 based on the assumption that the precipitation efficiency decreased with increased precipitation. Both equations follow the USDA Soil Conservation Service method (Smith, 1992; Allen et al., 1998), where effective rainfall is expressed as a fraction of total precipitation.

$$Pe = P \times (125 - 0.2P) / 125 \text{ for } P \leq 250 \text{ mm} \quad (10)$$

$$Pe = 0.1P + 125 \text{ for } P > 250 \text{ mm} \quad (11)$$

2.4 Random forest regression

Random Forest Regression (RFR) can be used to analyze interactions among variables and manage nonlinear relationships (Li et al., 2023). RFR builds multiple decision trees instead of a single one to reduce overfitting during training. It was used to predict *CWR* and *IWR* based on features (*CDI_Main*, *CDI_W*, *CDI_NCC*, month, and year) for selected river basins. The random forest was built with 500–1000 trees. We experimented with different settings and the number of features. Cross-validation was employed to ensure the model's generalizability. The model used bootstrapping, and to avoid overestimating accuracy, a rolling origin time split (80:20 ratio) was applied rather than random train-test splits. All data preparation steps, such as lagging and transformations, were performed within the training folds and then applied to the test folds. Model performance was assessed using multiple metrics, including mean squared error (MSE) and the coefficient of determination. Additionally, SHapley Additive exPlanations (SHAP) values were calculated to understand the contribution of features (Gebreyesus et al., 2023) to the model's predictions. SHAP decomposes each prediction to reveal how individual features influence the predicted *CWR* and *IWR*. Feature importance across all test folds was summarized, and the effects of *CDI* and its lags on *CWR* and *IWR*, including how their effects vary with seasonality, were examined.

3. Results

3.1 Spatial distribution of crops' relationships with drought, CWRs, and IWRs

The global distribution of major crops is presented in Appendix Fig. S1. Wheat and maize production were greater than rice productivity. Northeast China, North America, and Europe had high wheat production (more than 6 tons/hectare). In many parts of China, the United States, and Europe, wheat and maize are mainly grown under rainfed conditions. However, there are also significant irrigated areas, especially in northern China and

the western United States (Siebert et al., 2005). In contrast, rice in Southeast Asia and Sub-Saharan Africa depends on rainfed, and some parts of the United States, India, Japan, and China rely on irrigation. *CWR* and *IWR* were estimated for 110 major importers and exporters of cereals worldwide (Trade Map, 2023). Trade map. URL: <https://www.trademap.org> (Access Date: 3/3/2023)). The average monthly *CWR* were estimated for wheat (11.08 mm (low) to 41.80 mm (severe)), rice (35.89 mm (low) to 121.29 mm (severe)), and maize (19.36 mm (low) to 71.55 mm (severe)) global cropland areas. South China, North and South America, Africa, India, Southeast Asia, and Australia's northeast and southeast coasts presented higher *CWR*. Similarly, the average monthly estimated *IWR* were 0.62 mm to 7.10 mm for wheat, 7.55 mm to 34.92 mm for rice, and 2.75 mm to 14.80 mm for maize. The spatial distribution of the *IWR* area was lower than that of the *CWR* area. *IWR* was high in Pakistan, some parts of North America, South America, Africa, and Australia (Fig. 1). High and severe-dominated *CWR* and *IWR* clusters could be observed in significant cash crop countries.

Given the importance of cash crops, the composite indices were correlated with crop yields (Fig. 2 and Appendices S2 and S2.1). The indices showed a significant relationship with different crop yields. Wheat and maize yields showed a positive and significant correlation at the 0.05 level in North America, and a positive but nonsignificant correlation in European countries. In contrast, China, Australia, and South Asian countries dominated the mixed scenario (significant and non-significant relationships with drought indices). Rice had a negative correlation in most countries but a positive correlation in African countries. Rice is an irrigated and rainfed crop; thus, future droughts could reduce the yield of rainfed crops. The results indicate that composite drought indices exhibit stronger positive correlations with crop yields in major agricultural regions. However, negative correlation or lower correlation values were obtained in some areas, indicating that other factors also affect yield besides water availability in drought conditions. Findings revealed that composite drought indices could support a comprehensive drought impact assessment framework by integrating information that connects meteorological drought monitoring with hydrological and agricultural impacts. This could help with water resource planning and drought risk management.

The seasonal drought patterns and the positive relationship between drought and yield indicate that crop water requirements can be high during drought periods. *CWR* shows a positive correlation in most crop-producing countries. Among the drought indices, *CDI_W*, which assigns fixed weights to rainfall anomaly, water balance anomaly, and moisture index, exhibits stronger correlations (>0.5) in Australia, India, South America, and African countries. Conversely, *CDI_NCC*, which emphasizes current moisture conditions, and *CDI_Main*, which combines the three indices through PCA, show significant correlations mainly in Australia and scattered regions worldwide (Fig. 2; Appendix Fig. S2). These differences occur because crop yield is influenced differently by short-term moisture stress (better reflected by *CDI_NCC*), cumulative seasonal water balance (captured by *CDI_W*), and integrated multi-variable drought signals (represented by *CDI_Main*). The *IWR* relationships with the drought indices reveal very limited positive correlations (Appendix Fig. S3). For example, *CDI_W* shows positive but nonsignificant associations in parts of China, Australia, Africa, and Asia, while *CDI_Main* and *CDI_NCC* also display positive but nonsignificant

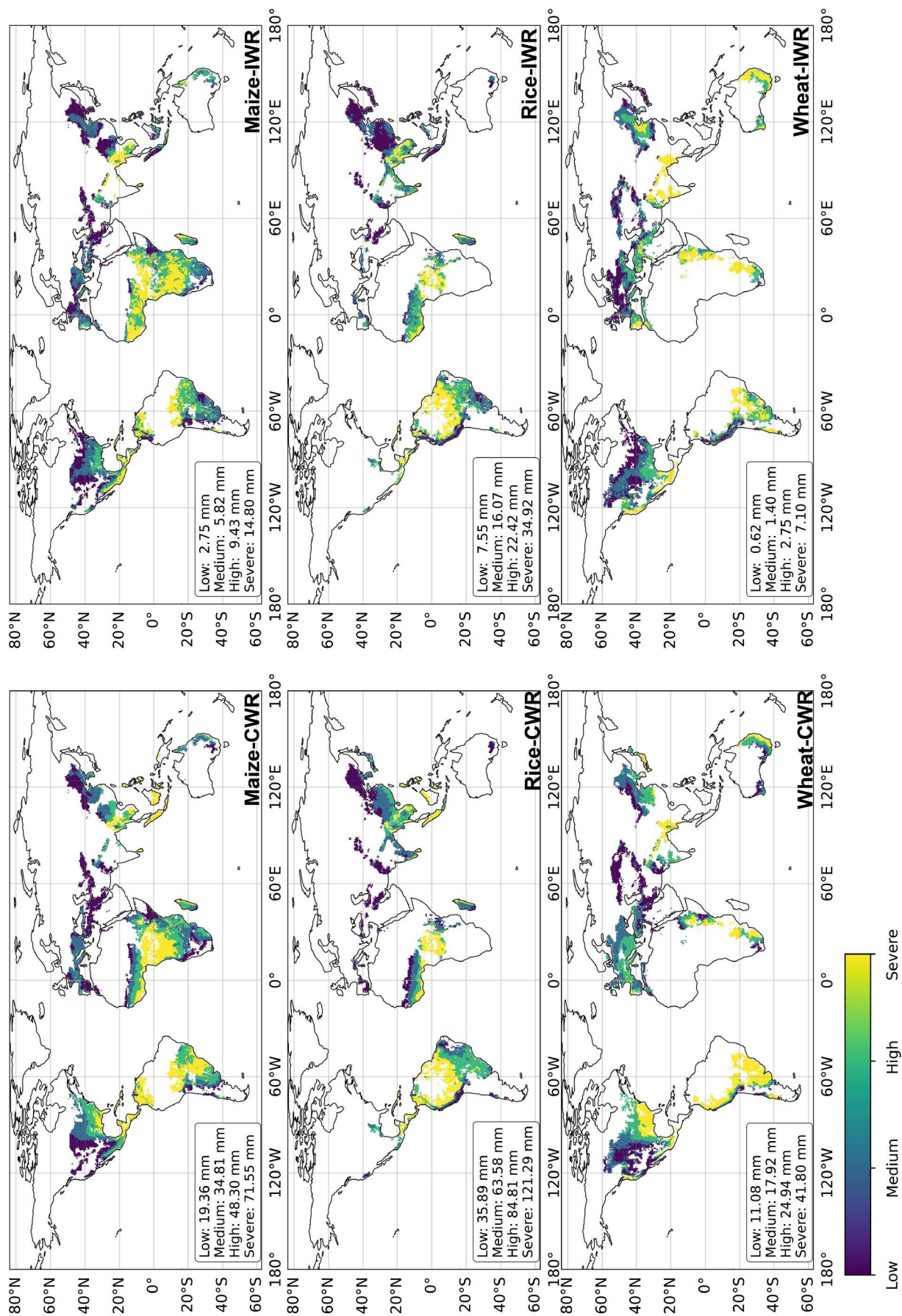


Fig. 1 Historical clustering (2000 to 2021) of CWR and IWR distributions for cash crops (maize, rice, and wheat). The clusters were created using thresholds computed from the 25th, 50th, 75th, and greater than 75th percentile, as well as the maximum value of the time-averaged data. Low, medium, high, and severe colors represent the distribution of water needed globally, based on the four corresponding clusters. For all plots, a typical color bar is used for comparison purposes.

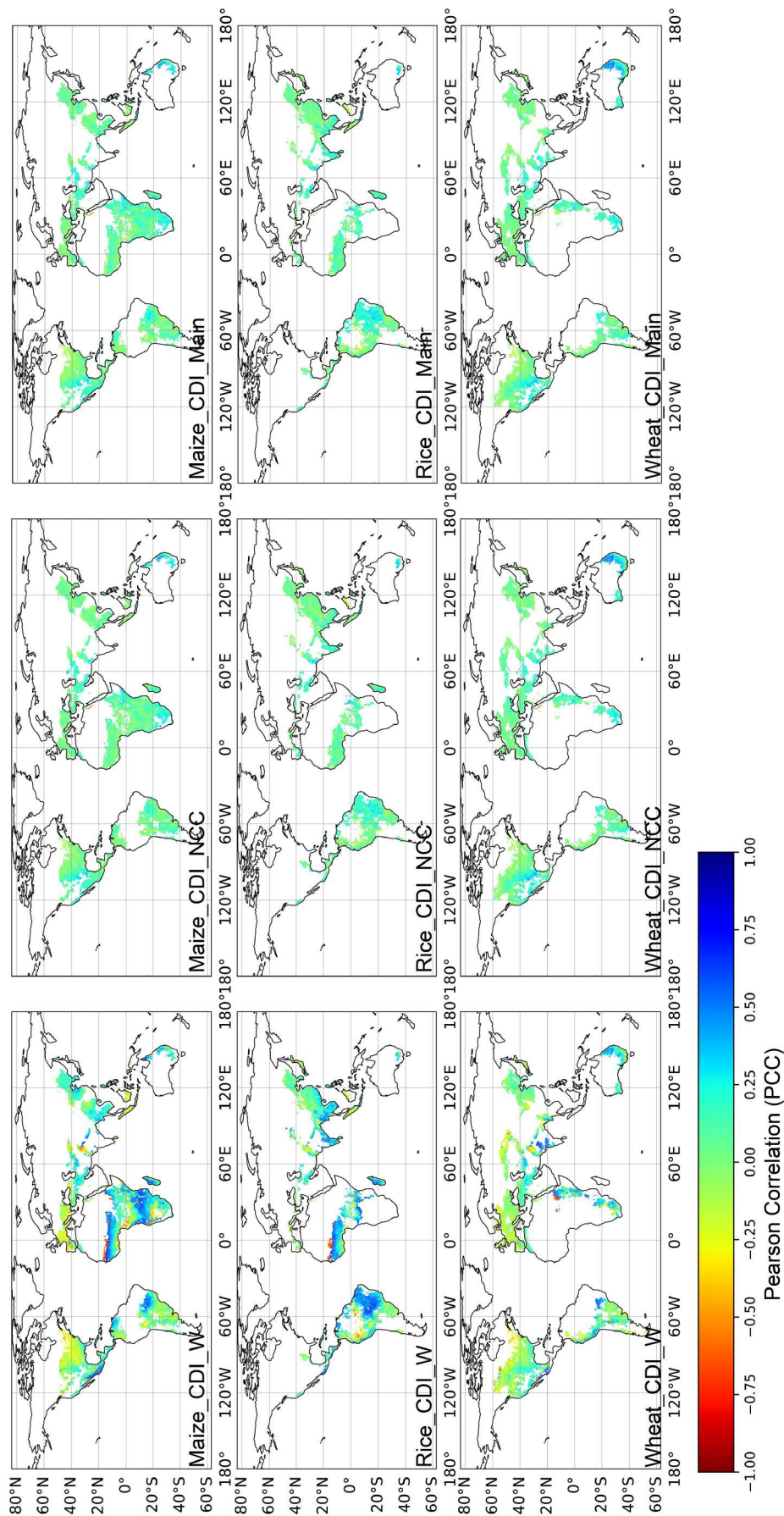


Fig. 2 Spatial distributions of correlations between composite indices and the CWRs of wheat, maize, and rice from 2000 to 2021. The correlations were calculated from the composite drought indices and the CWRs, showing the spatial pattern and correlation values over the global land. These plotted values are represented using a diverging color map, allowing for direct comparison of positive and negative values against each crop and drought index.

relationships. Overall, these variations highlight the complementary nature of the three *CDI* types: each emphasizes a different aspect of drought-crop interactions. Additionally, to establish a benchmark against widely used drought metrics, we also compared *IWR* relationships with the *SPI* and *SPEI* (Appendix Fig. S4). These comparisons showed that the *SPEI* was mostly negative, while the *SPI* exhibited positive relationships with rice *IWR* in some regions. The difference between *SPI/SPEI* and *CDI*-based indices highlights that traditional single-variable indices (*SPI*: precipitation-only, *SPEI*: *P-PET*) might not fully capture the complex interaction between drought stress and irrigation demand, especially where temperature-driven evapotranspiration or soil moisture storage has a larger impact. In contrast, the *CDI*, which combines multiple components (precipitation anomaly, water balance anomaly, and moisture index), reflects both supply and demand aspects of crop water needs. The generally weak or nonsignificant correlations between *SPI/SPEI* and *IWR* suggest that composite indices are more effective at explaining agricultural water stress. At the same time, discrepancies may also indicate improved irrigation management and crop adaptation to water shortages at maturity.

Given the above relationships, statistical metrics (R-squared, cross-validation (CV) R-squared, and MSE) indicated that the RFR model used to predict the impact of drought on *CWR* and *IWR* accounts for a significant portion of the variance. All three *CDI* forms (*CDI_W*, *CDI_NCC*, and *CDI_Main*) were tested; however, because *CDI_Main* combines multiple drought-influencing variables via PCA and provides the most consistent explanatory power across basins, it is emphasized here. The results for *CDI_W*, *CDI_NCC*, and *CDI_Main* are shown in Fig. 3, Appendix Figs. S5–S8. Results revealed that the composite drought indices, alongside temporal features, explain the variations in *CWR* and *IWR* well. The R-square values in the selected river basin mostly cross 0.9, suggesting the model is well applied to capture the variations. The SHAP plots were outlined for a more detailed analysis to highlight how composite indices influence *CWR* and *IWR* in the different river basins. The plots displayed that as the value of composite drought indices increased, the predicted *CWR* and *IWR* also increased. This relationship depends on regional climatic and agricultural variations across river basins. Results also describe the strong relationship between drought indices *CWR* and *IWR* during the growing season (indicated by warmer colors in plots), implying that sufficient water availability is crucial during critical growth periods. The geographical differences were also captured by SHAP values that indicate how composite indices influence *CWR* and *IWR*. For instance, the *CDI_Main* positive relationship, especially during the growing season in the Yellow River basin, revealed that water availability for maize is crucial during these months. The Indus and Sutlej exhibit a strong response to composite indices, indicating a dependency on monsoon rains and irrigation. Likewise, the Mediterranean Basin reflects the seasonal pattern and drought's influence on *CWR*, highlighting the importance of soil moisture and precipitation during the growing season. Similarly, the SHAP values revealed that composite indices explained mixed influences on *IWR* in the Yellow and Yangtze River basins. Low R-square and high MSE showed moderate prediction performance for maize, while for rice and wheat, the model performed well, suggesting an increase in *IWR* under water deficit conditions.

3.2 Projection under future change scenarios

We evaluated the composite drought indices for the spring and summer seasons under two SSPs (245 and 585) and nine models (described in the dataset and supporting information as supplementary material). Plots of dryness/wetness are presented in Appendix Figs. S9–S14. The climate models under both SSPs presented similar average dryness and wetness values from 2024 to 2060 and from 2061 to 2099, respectively. The first phase (2024 to 2060) had less dryness and more wetness, whereas the second phase (2061 to 2099) inferred more dryness worldwide. For example, North America, South America, Australia, Europe, and Asia presented high degrees of dryness. From 2023 to 2060, Asian countries, including North China, India, and Pakistan, experienced dryness in the spring season, while parts of Europe and Canada exhibited consistent wetness.

We also explored the composite drought index distributions under the various SSPs in the future to assess wetness and dryness in different basins. The average calculated indices were modeled for ten different river basins spanning the historical period (2000 to 2021) to the end of the century (2024 to 2099) under nine GCMs (Fig. 4 and Appendix Figs. S15 and S16). The variations in wetness and dryness can be seen as slightly greater than those historically reported. Under the SSP scenarios, the *CDI* distribution exhibited a slight negative skew, suggesting the potential for both dry and wet conditions in the future, with a tendency toward wetter conditions in certain regions. The distribution's peak was consistently near zero across all basins, pointing to dominant near-normal conditions compared to the historical baseline. In several basins, such as the Mississippi, Yellow, and Yangtze, there was evidence of a broader range of positive *CDI* values, implying a higher likelihood of wetter conditions ahead. The peak around zero needs careful interpretation. This pattern may partly stem from methodological factors, such as the limited historical reference period used in this study, which may restrict the ability to fully capture extreme drought and wet events. It also results from calculating *CDIs* values at the grid-cell level before aggregating over large basins. These aspects add some uncertainty to the distribution's shape and central tendency. However, the consistent trend of distributions broadening toward positive *CDIs* values across multiple models and scenarios suggests that the anticipated increase in wetness is not merely due to methodological or dataset biases but is a robust finding from ensemble projections. This underscores the need to account for both methodological uncertainty and multi-model agreement when analyzing future drought and wetness trends (a detailed explanation is provided in Appendix S2.2). Furthermore, it is worth noting that the different *CDI* variants (*CDI_W*, *CDI_NCC*, and *CDI_Main*) exhibited somewhat different dryness and wetness distributions. This variation reflects their different sensitivities to the calculation or formulation used to calculate the moisture index. These differences highlight that future drought assessments depend on the index used, underscoring the need to interpret results within the broader, multidimensional context of drought processes.

3.3 Effect on water requirements

The percentage change in *CWR* and *IWR* was estimated under the SSPs for historical periods (2000 to 2016) and for future periods (2024 to 2060 and 2061 to 2099) (Appendix Figs. S17

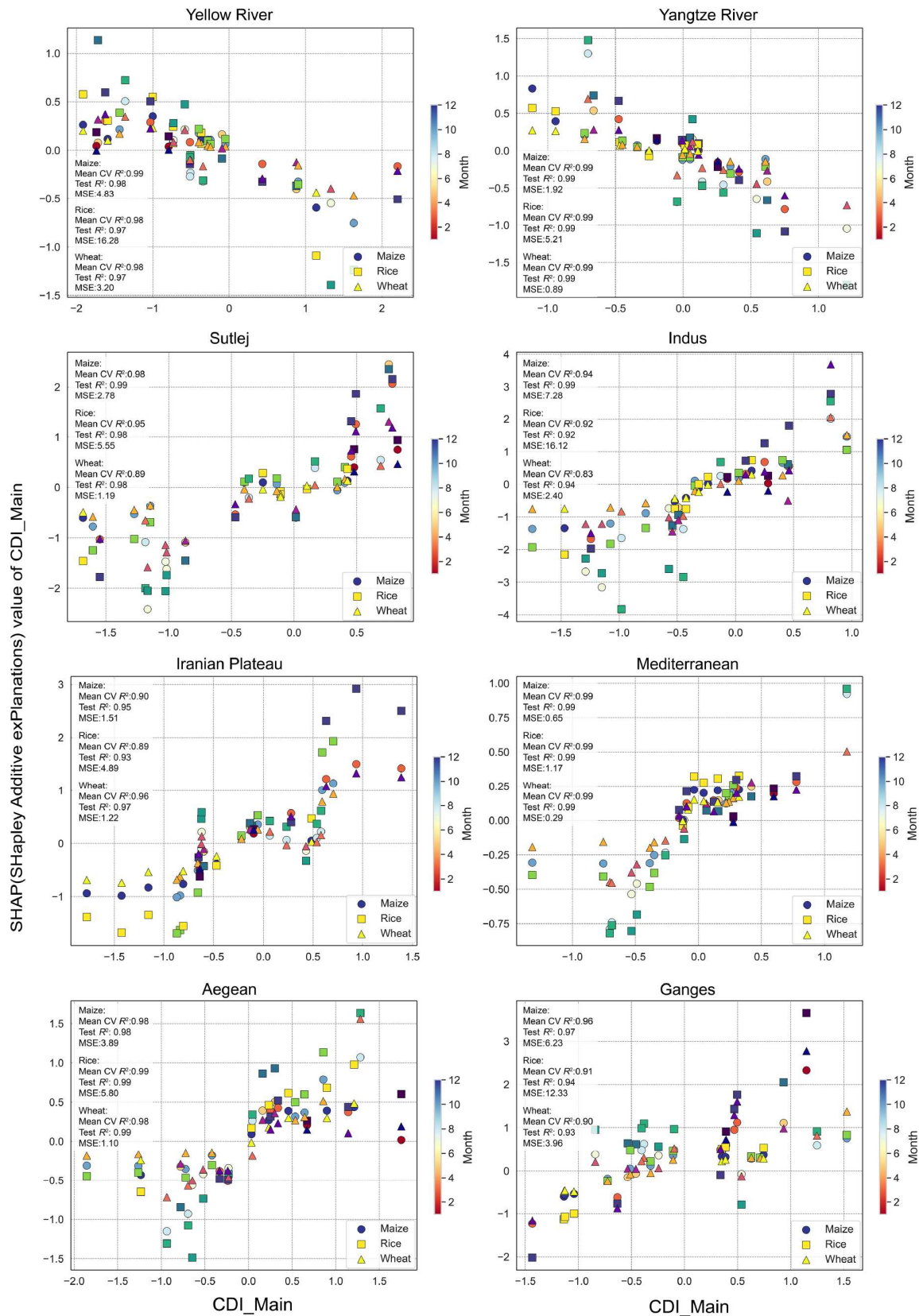


Fig. 3 CDI_Main relationship with CWR of maize, wheat, and rice crops using random forest regression and SHAP analysis. Each subplot shows a particular basin, indicating the variability of CWR for maize, rice, and wheat to CDI_Main for a period (2000 to 2021). SHAP dependence plots illustrate how CDI_Main impacts CWR; the color of the markers indicates the month of the year, ranging from January to December. This is done to enable the observation of seasonality patterns in the relationship between CDI_Main and CWR. The basin plots are labeled with mean cross-validation R^2 and MSE.

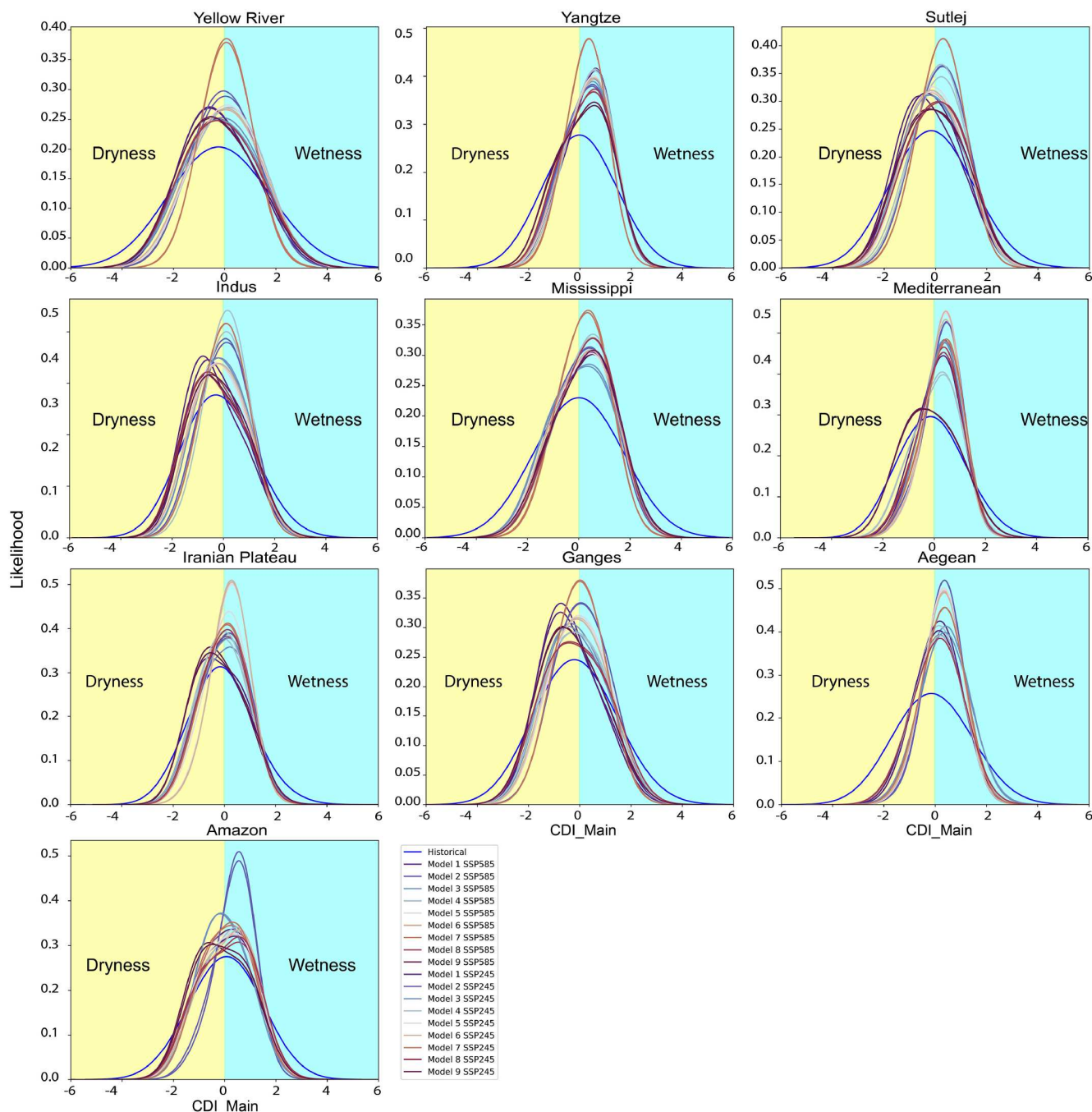


Fig. 4 *CDI_Main* distributions in ten selected river basins across the globe. *CDI_Main* distributions are plotted for historical (2000 to 2021) and nine SSPs (285 and 585) periods (2023 to 2099). Each subplot corresponds to a river basin and compares the shift in the probability of dry and wet extremes. Kernel Density Estimation (KDE) plots the probability density of *CDI_Main*. Cyan-colored areas indicate wetness values, and yellow-colored areas represent dryness values.

and S18). In general, there was an increase in wheat and maize CWR, greater than 30% under both SSPs. Most European countries had increased CWR (greater than 50%), while South and North America, as well as China, also experienced significantly increased CWR. For rice CWR, most parts of South America experienced a decrease in CWR, but some areas showed more than a 75% increase in CWR. European countries, China, and African nations where rice is cultivated presented significant increases in CWR. In contrast, some parts of China and North and South America presented a rise in IWR, whereas most

regions presented a decrease in IWRs (Appendix Figs. S19–S20). Results revealed a strong relationship between IWR and CWR with drought; however, many areas are independent of extreme events. In those regions, water requirements appear to be less directly affected by extreme events, likely due to the buffering effects of irrigation management or regional hydrological conditions. For example, in European countries, certain regions, like France, face a dual challenge: more frequent severe droughts, as seen in 2003 and 2022 (Bevacqua et al., 2024), and more intense floods caused by warming-triggered heavy rainfall, such

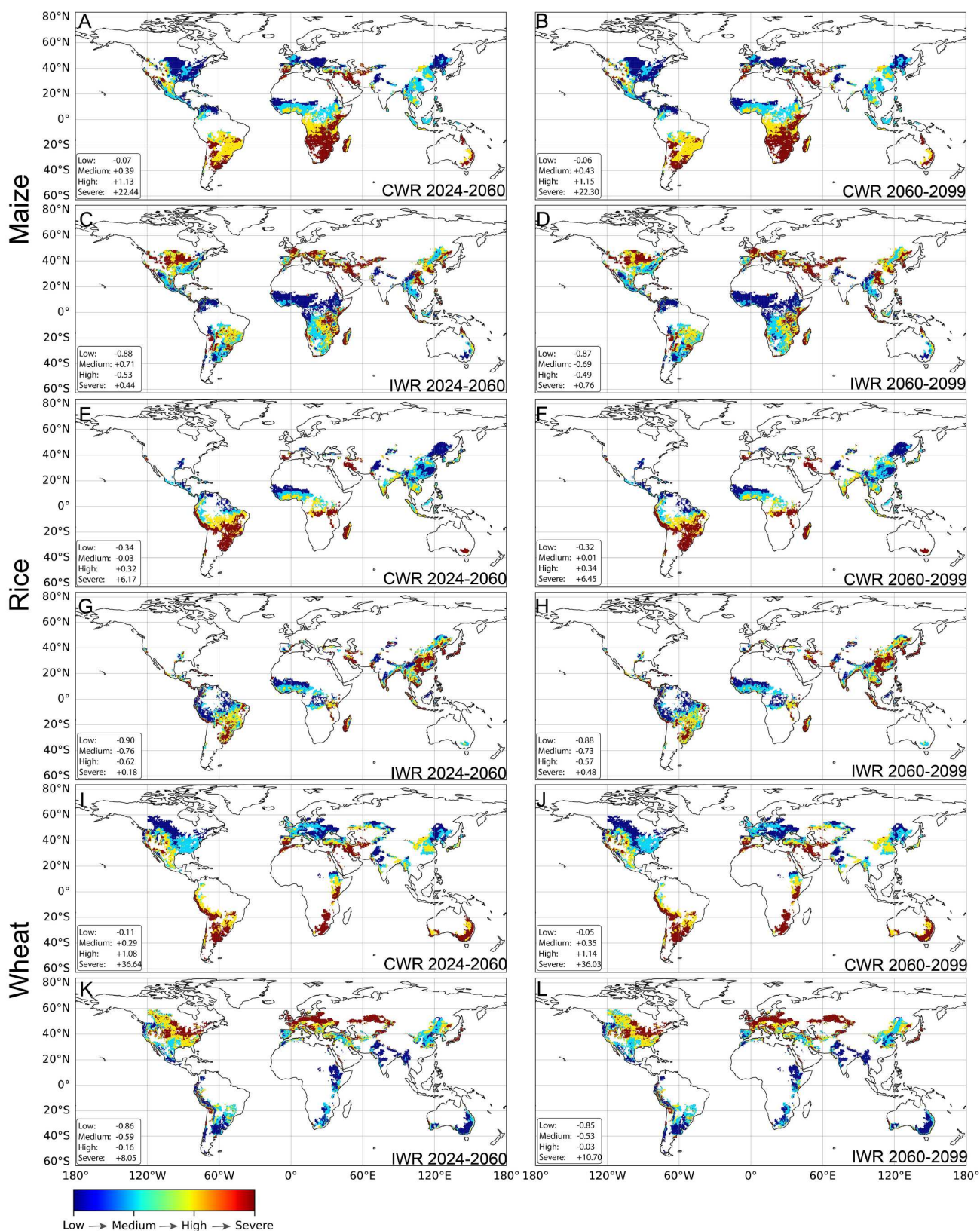


Fig. 5 Future clustering of total (including rainfed and irrigated land) CWR and IWR demand distributions for maize, rice, and wheat crops using ensembles of General circulation model (GCMs) under SSP 585 at 99th percentiles. Coloring low, medium, high, and severe represents the distribution of water requirements. CWR for maize subplots panel (A, B) from 2024 to 2060 and 2061 to 2099; subplots panel (C, D) represents IWR for maize; CWR for rice subplots panel (E, F) from 2024 to 2060 and 2061 to 2099; subplots panel (G, H) represents IWR for rice; CWR for wheat subplots panel (I, J) from 2024 to 2060 and 2061 to 2099; subplots panel (K, L) represents IWR for wheat. The legend labeled with low, medium, high, and severe represents clusters' mean CWR and IWR values.

as the 2020 Alpine floods and the doubled chance of floods in Paris (Kouggoulos et al., 2021; Trambly et al., 2025), suggesting that water demand and availability are not always aligned with a single drought or flood extremes. Moreover, the CWR and IWR clusters showed France was listed under medium to high water requirements (Fig. 5, Appendix Figs. S21–S25). Asian and Middle Eastern countries, as well as some parts of South America and Africa, presented high IWR and CWR in both periods under the SSPs. The fractional change in CWR and IWR per month relative to the baseline period was estimated for wheat (SSP585: +0.09 mm to +80.16 mm; SSP245: +0.53 mm to +39.12 mm) and for (SSP585: +0.55 mm to +42.68 mm; SSP245: +0.52 mm to +39.12 mm), respectively. Results exhibited that the fractional change in CWR for rice is expected to increase from −0.09 mm/month to +14.01 mm/month. This suggests that a substantial future demand for agricultural water is necessary for crop production, which may pose a significant regional challenge to agricultural systems.

4. Discussion and summary

We used composite indices to demonstrate the effects of wetness and dryness on CWR, IWR, and crop yields across different basins and global scales. For a robust analysis, the evaluated datasets were employed to account for the contributions of climate warming. SPI and SPEI were used for comparison purposes, and the SPEI calculation was made using precipitation and GLEAM actual evaporation data. The results reaffirm the CDI's ability to represent drought conditions across regions worldwide effectively. However, it is essential to note that the previous study (Faiz et al., 2022) was validated using only one region. The differences observed between CDI, SPI, and SPEI can be explained because CDIs incorporate multiple variables (precipitation, evapotranspiration, and climatic moisture deficit), making it comparable to capturing and evaluating drought effects on agricultural systems. Although SPI and SPEI are useful for detecting precipitation and water balance anomalies, our case study comparisons show that CDIs can reveal additional spatiotemporal differences, especially in climate transition zones or semi-arid areas. Overall, the results of CDIs demonstrate that CDIs can enhance water management and agricultural productivity under changing climate conditions. This is especially important in rain-fed agricultural regions that are highly sensitive to drought characterization for ensuring food security. Moreover, the choice of data can lead to different estimates of drought characteristics (Beguería et al., 2014), which in turn influences assessments of agricultural vulnerability and food security. Previous studies (Zhao et al., 2017; Solaraju-Murali et al., 2021) have demonstrated that drought and heat stress significantly reduce crop yields at both global and regional levels. By integrating multiple drought drivers, the CDI framework could be better equipped to capture such combined stresses, thereby offering reliable insights into agricultural risks under future climate scenarios. Composite indices represented well the European summer drought in 2018, as recorded by previous studies (Blauhut et al., 2022; Rakovec et al., 2022). An increase in the CWR and IWR of selected crops was dominant in some regions and may affect food security. Anthropogenic influences have greater effects on precipitation variability, such as in Europe (Christidis and Stott, 2021), which could exacerbate this phenomenon.

The results provide vital features of CWR and IWR changes and the effects of drought on crop yield (similar findings reported by (Marcos-Garcia et al., 2024) that crop risk is associated with dry spells at the regional level). We found a 30% to 75% increase in CWR at a regional level, which could be compensated by irrigation; regionally, up to 50% of yield could be lost under severe drought (Zwart and Bastiaanssen, 2004). Although the CDIs, IWR/CWR, are statistical in nature, they aim to represent biophysical processes that restrict crop growth and productivity. Drought indices track the imbalance between water supply and atmospheric moisture demand. When soil water is scarce, stomatal closure cuts back transpiration and CO₂ uptake, which directly limits photosynthesis and biomass production (Lobell and Gourdj, 2012; Zhang and Cai, 2013). Soil moisture deficits and rooting depth affect crop resilience. Reduced rainfall and increased evaporative demand lower soil moisture, especially in the root zone. Crops with shallow roots (e.g., rice, wheat) are more susceptible to short-term soil water shortages. In contrast, deep-rooted crops (e.g., maize) can temporarily endure stress but are vulnerable under prolonged deficits (Sinclair and Muchow, 2001). Phenological sensitivity affects yields, particularly when drought occurs during key growth stages, such as flowering and grain filling, resulting in incomplete pollination, lower grain set, and a reduced harvest index (Barnabás et al., 2008). In irrigated systems, IWR increases during drought. If the water supply cannot meet this increased demand due to surface water shortages or groundwater depletion, yield losses happen despite irrigation efforts (Siebert and Döll, 2010). We also acknowledge that CWR and IWR estimation on a larger scale would be a helpful extension of our research. The results could be promising, as we considered only composite indices for future estimations. For reliable prediction or variation estimation, a large ensemble is needed (Smith et al., 2019). To achieve this, heat stress, excess water, and climate oscillations could be examined in relation to yield, CWR, and IWR variations. The results of this study can be considered a first step in the development of climate services in crop regions. Simple composite indices could be used to support and provide drought information to farmers and crop breeders. They can benefit from drought information when deciding crop diversification and drought tolerance for their breeding programs. Overall, the present findings identify the need to take action to better manage agricultural water use.

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Supporting information

The supporting information is available online at <http://earth.scichina.com> and <http://link.springer.com>. The supporting materials are published as submitted, without typesetting or editing. The responsibility for scientific accuracy and content remains entirely with the authors.

Data availability

Precipitation input data for calculating the Composite Drought Index are available at (www.gloh2o.org/mswx), and potential and actual evaporation datasets are available at (<https://doi.org/10.5281/zenodo.7789759>) and (<https://www.gleam.eu/#datasets>). CMIP6 model data is available at (<https://cds.climate.copernicus.eu/cdsapp#!/dataset/projections-cmip6?tab=form>); crop yield data were obtained from (<https://doi.pangaea.de/10.1594/PANGAEA.909132>). Code availability: No specific code or software was used for data collection; for data analysis and plotting, Python 3.12.3 libraries are used: “netCDF4”, “pandas”, “numpy”, “scipy.interpolate”, “matplotlib.pyplot”, “xarray”, “geopandas”, “rioarray”, “cartopy.crs”, “cartopy.feature”.

“sklearn.ensemble”, “sklearn.metrics”, “sklearn.model_selection”, “RandomForestRegressor”, “climate-indices”.

Compliance and ethics

The authors declare no conflict of interest.

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